



# Vojtěch Špaček

**Senior C++ engineer — safety-critical embedded & HIL test automation | Linux/networking | agentic-AI (MCP) tooling**

*Pragmatic programmer and creative problem solver — systems, verification and open source.*

Location: Brno, CZ | Citizenship: EU (Czech Republic)

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Signal: [signal.me/#eu/-i6EcG8dz6FtV6U2yytXP4UESGjNMxGTw4Xf8AWbR3b5dD3lyBKd53IEKi5AJ6N](https://signal.me/#eu/-i6EcG8dz6FtV6U2yytXP4UESGjNMxGTw4Xf8AWbR3b5dD3lyBKd53IEKi5AJ6N) |

YouTube: [youtube.com/playlist?list=PLaWOvDBjg6WiUcQm-yEP1RskMfPeWMKTL](https://youtube.com/playlist?list=PLaWOvDBjg6WiUcQm-yEP1RskMfPeWMKTL) | Discord:

[discord.com/channels/@sparesparrow](https://discord.com/channels/@sparesparrow) | Website: [sparesparrow.github.io/cv/](https://sparesparrow.github.io/cv/)

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## Profile Summary

Senior C++ engineer working on safety-critical and real-time systems: aircraft onboard diagnostics at Honeywell, electron-microscopy detector software at Thermo Fisher Scientific, and a Linux inspection proxy and firewall at Trusted Network Solutions.

Most recently sole owner of the hardware-in-the-loop verification suite and automated bench for a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), raising measured conformance to 259 of 271 tracked state transitions (95.6%).

Linux, networking and security specialist — Netlink, nftables, TLS inspection, Kerberos/LDAP/OCSP — with C++20 daemons shipped as Debian packages and a Rust reimplementations published on crates.io.

Builds agentic-AI tooling on the Model Context Protocol: an agent harness adopted into a daily client delivery workflow, and MCP-Prompts, published to npm, Docker Hub and the MCP directories (Glama, MCP Market, PulseMCP) with 97 stars and 21 forks.

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## Work Experience

### Test Automation Engineer (Python, C#)

Resideo (Ademco) (via CoolPeople (contractor))

Hardware-in-the-loop verification of a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), as sole owner of the verification test suite and of the automated bench that drives it. Owned delivery end to end: requirement traceability, issue tracking, branch and pull-request workflow, Conan packaging, CI integration and monthly client reporting.

- **Engineered the bench device layer so each instrument is owned by a single thread against a shared monotonic clock, covering a pyvisa/USBTMC/SCPI programmable AC source, CAN serial-bus monitoring and Modbus relay and probe I/O.**
- **Raised measured conformance to 259 of 271 tracked transitions (95.6%) and closed a ~148-case gap in which stimulus windows were never actually measured, bringing it to zero.**
- **Investigated 25+ firmware and harness defects against the C/C++ firmware sources, shipping a standalone reproduction script for each.**

### Software Developer (C++)

Thermo Fisher Scientific (via Ixperta (contractor)) · 2023 – 2024

Electron microscopy software for high-resolution detector hardware operation and image acquisition. Built server-side C++ applications, FPGA-based data processing pipelines with gRPC networking, and backend logic for a C#/NET WPF image processing and display application. Worked in an international Agile team with bi-weekly stakeholder demos.

- **Implemented server-side software in C++, including hardware abstraction layers (HAL) and inter-process communication (IPC) using COM and MIDL on Windows.**
- **Developed embedded software for FPGA hardware, using gRPC network communication in C++, Python and Linux environments.**
- **Optimized the data acquisition pipeline and implemented robust error handling.**

Software Engineer II

Honeywell · 2022 – 2023

Onboard Maintenance System that monitors the condition of aircraft and diagnoses issues during flight. Architected and implemented the DiagTest framework, a state machine-based diagnostic test engine in C++ managing 15+ operational states for aircraft member system testing during flight and maintenance operations.

- **Designed a job-based execution system with DiagTestJob and DiagTestMsgQueProc for asynchronous test orchestration, implementing message queue processing patterns for reliable real-time diagnostics.**
- **Engineered a test interference management system preventing conflicting diagnostic tests from running simultaneously, ensuring aircraft system safety and data integrity.**
- **Implemented four diagnostic test modes (data collection, passive monitoring, interactive, non-interactive) with timeout management, precondition handling and aviation-standard command/response protocols.**

Skills

Embedded & RTOS

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|---------------------------------------------------------------------|-------------|
| C++ (C++17/C++20) (Senior)                                          | <div></div> |
| Embedded software / ARM / RTOS (Medior)                             | <div></div> |
| State machine design patterns (Senior)                              | <div></div> |
| Hardware abstraction layers & IPC (COM/MIDL, Unix sockets) (Medior) | <div></div> |
| FPGA-adjacent development & gRPC (Medior)                           | <div></div> |
| Flatbuffers (binary serialization) (Medior)                         | <div></div> |
| Qt/QML (Working knowledge)                                          | <div></div> |

Networking & Security

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|-----------------------------------------------------------|-------------|
| Linux (Senior)                                            | <div></div> |
| Linux networking, TCP/IP, Netlink, nftables (Senior)      | <div></div> |
| Encrypted traffic inspection (TLS 1.3, OCSP) (Medior)     | <div></div> |
| Kerberos/Heimdal, LDAP, PostgreSQL, Suricata IDS (Medior) | <div></div> |
| Forward/TCP/HTTP proxy design (Senior)                    | <div></div> |
| Rust (Working knowledge)                                  | <div></div> |

Tooling

|                                                               |             |
|---------------------------------------------------------------|-------------|
| Git (Senior)                                                  | <div></div> |
| Bash (Senior)                                                 | <div></div> |
| Docker / Podman (Medior)                                      | <div></div> |
| Conan (C/C++ dependency management) (Medior)                  | <div></div> |
| CI/CD (Jenkins, TeamCity, GitLab CI, GitHub Actions) (Medior) | <div></div> |
| PowerShell (Medior)                                           | <div></div> |
| Enterprise Architect (UML) (Working knowledge)                | <div></div> |
| GDB, Wireshark, VirtualBox (Medior)                           | <div></div> |

Verification & HIL

|                                                                                       |             |
|---------------------------------------------------------------------------------------|-------------|
| Python (Medior)                                                                       | <div></div> |
| Hardware-in-the-loop test design (Senior)                                             | <div></div> |
| pyvisa / USBTMC / SCPI instrument control (Medior)                                    | <div></div> |
| Modbus I/O cards & CAN serial-bus diagnostics (Medior)                                | <div></div> |
| Safety-certified product verification (UL/CSA, ANSI Z21.20, NFPA 54) (Medior)         | <div></div> |
| Requirement traceability & test evidence reporting (openpyxl, HTML, Mermaid) (Medior) | <div></div> |
| C# / .NET (Medior)                                                                    | <div></div> |
| Perl test frameworks (Medior)                                                         | <div></div> |

AI / LLM / MCP

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|-----------------------------------------------------------------------------------------------|-------------|
| Model Context Protocol server development (Senior)                                            | <div></div> |
| Agentic workflow design & governance (agents, skills, policy hooks) (Medior)                  | <div></div> |
| Prompt engineering (RAG, few-shot, prompt caching, structured outputs, tool calling) (Medior) | <div></div> |
| LLM provider APIs & SDK integration (Medior)                                                  | <div></div> |
| JavaScript / TypeScript (Medior)                                                              | <div></div> |
| Node.js (Medior)                                                                              | <div></div> |

Availability & Work Authorization

Available as an independent contractor (Czech IČO). Brno on-site and remote. Trading on a Czech IČO. EU citizen (Czech Republic) — no permit required in the EU/EEA.

Languages

Czech (native) · Slovak (native) · English (C1) · German (A1)